The relation set of `Simplified-V1.Syntactics`

`Clifford-Relations.Base._SRel,_===_` together with the structural rules of `Circuit.Base.Lift-Relation`.

Diagrams read left to right in matrix-multiplication order (the *rightmost* gate is applied first), so each transcribes its Agda word letter for letter. Wires are numbered from the bottom: w sits on wire 0, $w\uparrow$ on wire 1, and $w\downarrow$ is w itself.

Derived words (definitions)

Z

$$Z \equiv H\; H\; S\; H\; H\; S^{p-1}$$

X

$$X \equiv H\; S\; H\; H\; S^{p-1}\; H$$

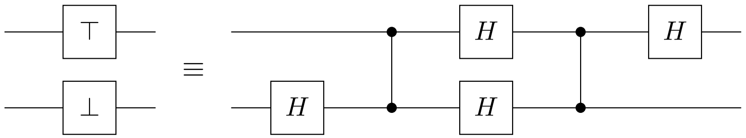
R

$$R \equiv S\; Z^{1/2}$$

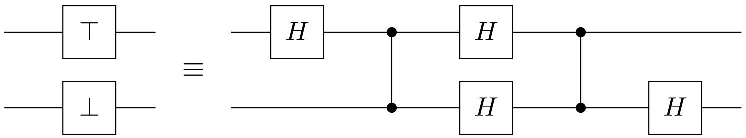
M_x

$$M_x \equiv R^x\; H\; R^{x^{-1}}\; H\; R^x\; H \qquad x \neq 0$$

$\perp\top$



$\top\perp$



One-wire axioms

order-S

$$S^p \equiv \text{—}$$

order-H

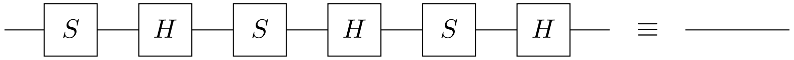
$$H\; H \equiv M_{-1}$$

M-power

$$(M_g)^k \equiv M_{g^k}$$

semi-MR

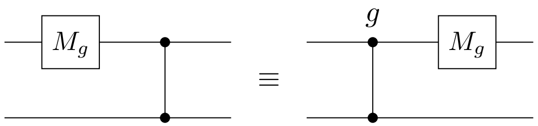
order-SH



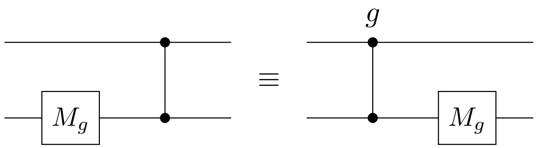
comm-HHSHHS

Two-wire axioms

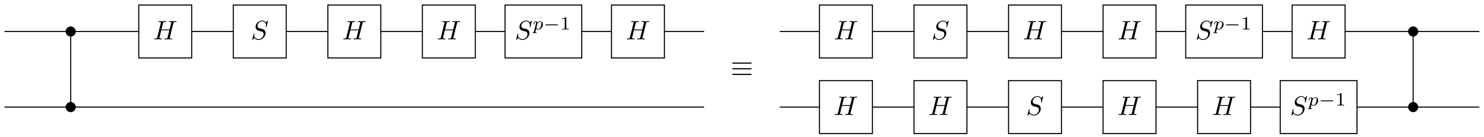
semi-M↑CZ



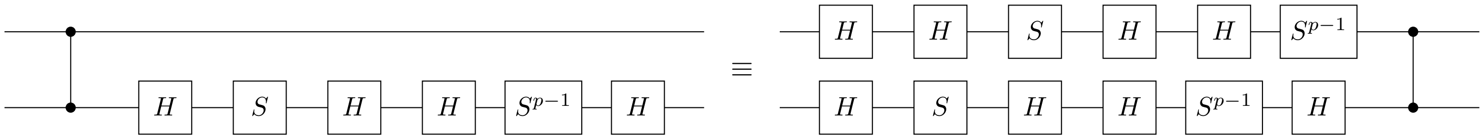
semi-M↓CZ



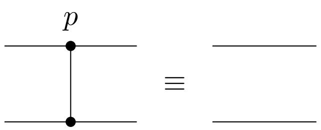
rel-X↑-CZ



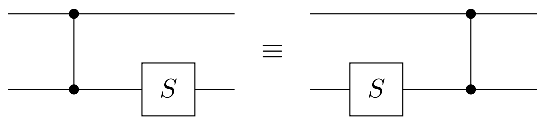
rel-X↓-CZ



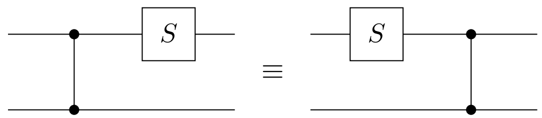
order-CZ



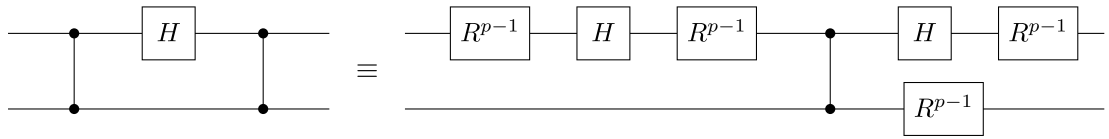
comm-CZ-S↓



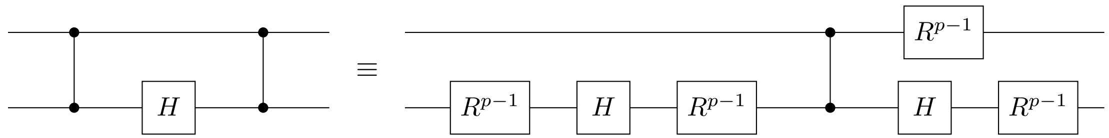
comm-CZ-S↑



selinger-c10

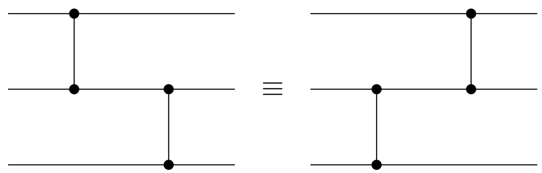


selinger-c11

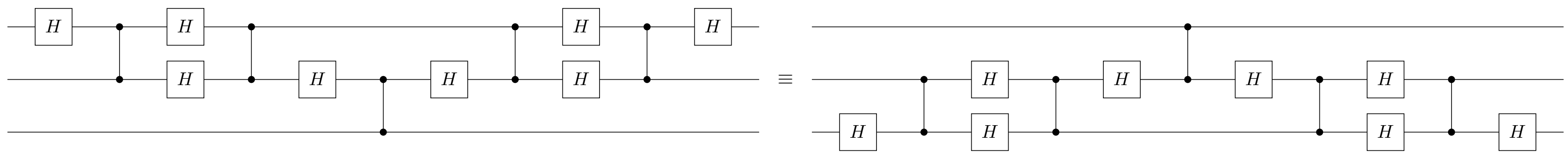


Three-wire axioms

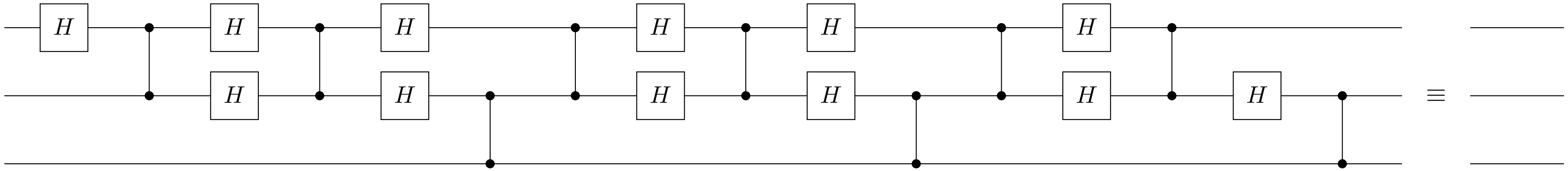
selinger-c12



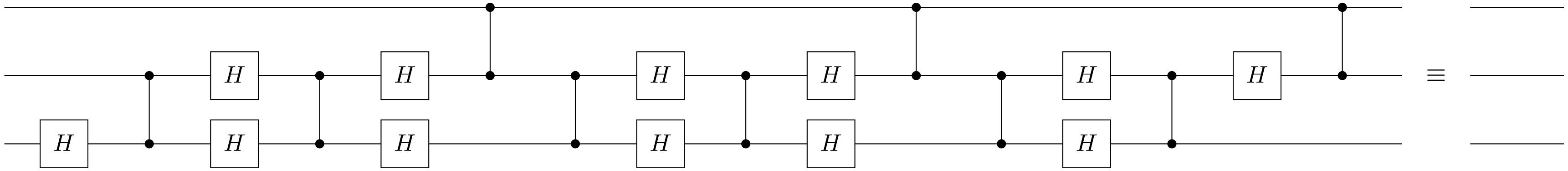
selinger-c13



selinger-c14

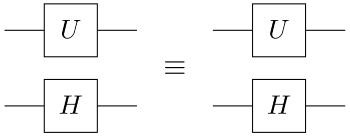


selinger-c15

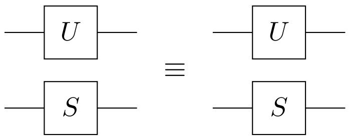


Structural rules (Circuit.Base.Lift-Relation)

comm-H



comm-S



comm-CZ